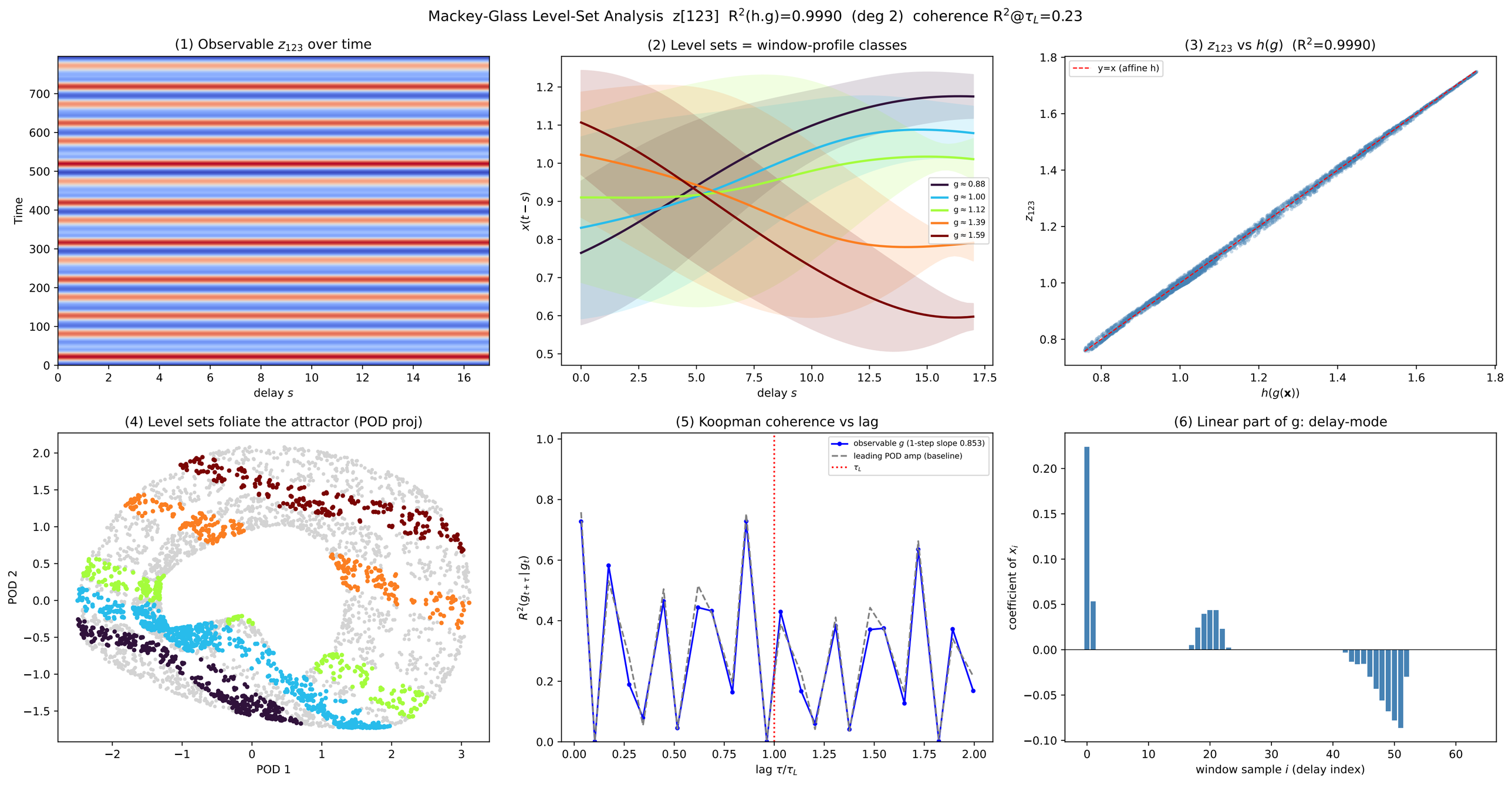

Mackey-Glass Level-Set Analysis z[123] R²(h.g)=0.9990 (deg 2) coherence R²@τ_L=0.23
(1) Observable z_123 over time
Time
delay s
(2) Level sets = window-profile classes
x(t − s)
delay s
g ≈ 0.88
g ≈ 1.00
g ≈ 1.12
g ≈ 1.39
g ≈ 1.59
(3) z_123 vs h(g) (R²=0.9990)
y=x (affine h)
z_123
h(g(x))
(4) Level sets foliate the attractor (POD proj)
POD 2
POD 1
(5) Koopman coherence vs lag
observable g (1-step slope 0.853)
leading POD amp (baseline)
τ_L
R²(g_{t+τ} | g_t)
lag τ/τ_L
(6) Linear part of g: delay-mode
coefficient of x_i
window sample i (delay index)